

SML 2026, Winter, Quiz 1, Dur. 1 hr 10 mins. Total marks 10

Instructions

Q1 to Q4 are for all.

Only for CSE542/ECE556. Q5 and Q6 are only for CSE542/ECE556. The marks will be scaled out of 10.

Q1. Let $x \in \mathbb{R}^2$. Assume

$$p(x | \omega_1) = \mathcal{N}(\mu_1, \Sigma), \quad p(x | \omega_2) = \mathcal{N}(\mu_2, \Sigma),$$

with equal priors. The loss matrix is

$$\Lambda = \begin{bmatrix} 0 & \lambda_{12} \\ \lambda_{21} & 0 \end{bmatrix}, \quad \lambda_{12} = 1, \lambda_{21} = 1.$$

Given

$$\mu_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mu_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

1. Find the likelihood ratio decision rule and show it can be written as a linear test

$$w^\top x > \tau.$$

Give w and τ explicitly. [1.5][CO1]

2. Classify [.5] [CO1]

$$x = \begin{bmatrix} 0.4 \\ 0.2 \end{bmatrix}.$$

Q2. Consider a binary classification problem with feature $x \in \mathbb{R}$. The class-conditionals are:

$$p(x | \omega_1) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad (\text{Gaussian } \mathcal{N}(0, 1)),$$

$$p(x | \omega_2) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0, \end{cases} \quad \text{where } \lambda = \frac{1}{\sqrt{2\pi}}.$$

Assume equal priors $P(\omega_1) = P(\omega_2) = \frac{1}{2}$ and 0-1 loss.

1. Derive the Bayes decision rule and find that the decision boundary. You must clearly specify the decision regions on $\mathbb{R}$. [2] [CO3]
2. Using your decision regions, compute the Bayes probability of error. You can give it in the form of $\text{erfc}(\cdot)$. [2] [CO3]
3. Let the probability of misclassifying class ω_1 (Gaussian) as class ω_2 (Exponential) be:

$$P(\text{decide } \omega_2 | x \in \omega_1).$$

Suppose you want to design a threshold rule that leads to $P(\text{decide } \omega_2 | x \in \omega_1) < E_1$. Derive the decision boundary (give the threshold explicitly in terms of E_1), and clearly state the resulting decision regions. In case you arrive at an equality using $\text{erfc}(\cdot)$, you can leave the answer in that equality form. [1] [CO3]

4. For the decision boundary obtained in part (c), give an expression the following error probability [1] [CO3]

$$P(\text{decide } \omega_1 \mid x \in \omega_2)$$

Note, you can evaluate this integral explicitly and leave in the form of decision boundary.

Q3. Assume 0-1 loss. Under what conditions do minimax risk and Bayes risk give the same decision boundary. Explain using an example. [1] [CO1]

Q4 Consider binary classification with a scalar feature $x \in \mathbb{R}$ and class-conditionals

$$p(x \mid \omega_1) = \mathcal{N}(\mu_1, \sigma^2), \quad p(x \mid \omega_2) = \mathcal{N}(\mu_2, \sigma^2),$$

where $\mu_2 > \mu_1$ and σ^2 is known. Assume 0-1 loss and equal priors.

Let P_e^{Bayes} denote the Bayes error, and let P_e^{NP} denote the overall probability of error for the Neyman–Pearson decision boundary x^* .

Is $P_e^{Bayes} \leq P_e^{NP}$ in general? Using a sketch of the two Gaussian PDFs, explain why. When does the equality hold? [1][CO1]

Following are only for CSE542/ECE556

Q5. Consider a binary classification problem with posterior

$$\eta(x) := P(\omega_1 \mid x) \in [0, 1].$$

Two candidate upper bounds for Bayes error are:

$$B_1(x) = \eta(x)^{1-\eta(x)} (1 - \eta(x))^{\eta(x)}, \quad B_2(x) = \sqrt{\eta(x)(1 - \eta(x))}.$$

(a) Between $B_1(x)$ and $B_2(x)$ which is a tighter bound? Show the proof. [1] [CO3]

Q6. Give an integral expression for $P(\text{error})$ considering the setting in Figure 1. Assume equal priors. You must clearly give the limit values. You do not need to solve the integral. [1][CO3]

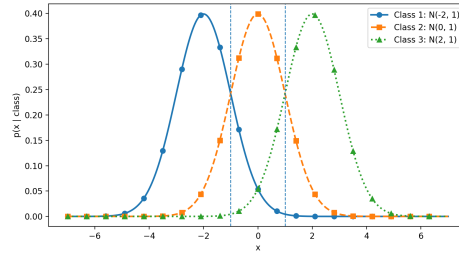


Figure 1: Class conditionals